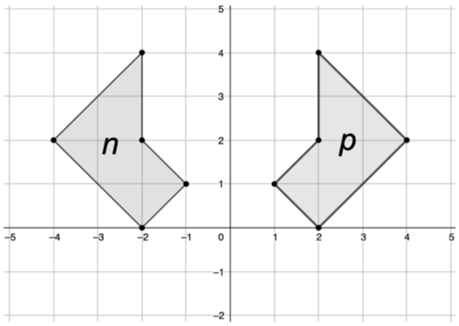
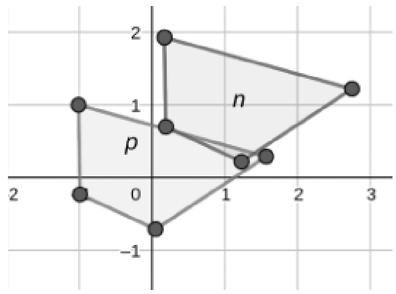
Grade 8 Name_Answer Key________
Unit 9 Date____________________
Lesson 6 Practice Period___________________

On the coordinate grids below, figure n is the result of a transformation on figure p.

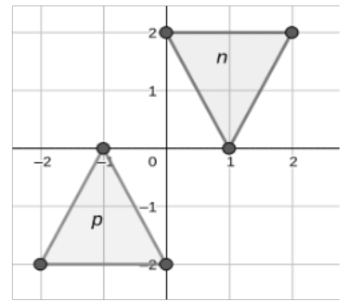
1.



2.



3.



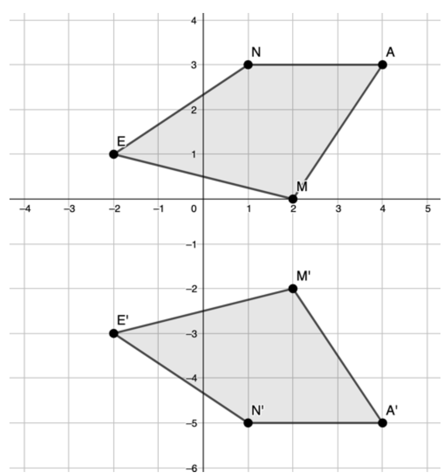
Match the grid to the statement that best describes the transformation. Then complete the chosen statement.

__2__ figure p was translated __1__ unit(s) [o left / o right] and __1__ units [o up / o down] to map to figure n.

__1__ figure p was reflected to map to figure n. The line of reflection is the y-axis______.

__3__ figure p was rotated about the origin to map to figure n. The angle of rotation is __180__°.

Sarah, Diego and Jayden are discussing the image to identify what single transformation took place to map Quadrilateral $NAME$ to Quadrilateral $N'A'M'E'$.

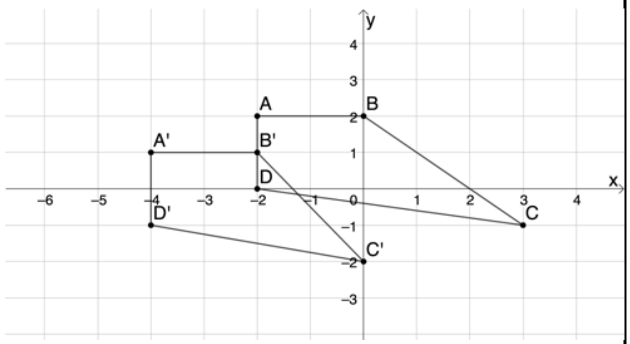
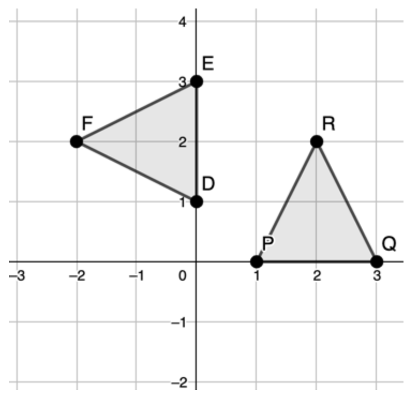
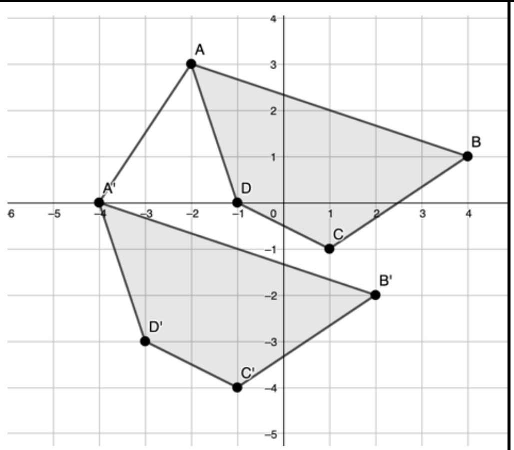
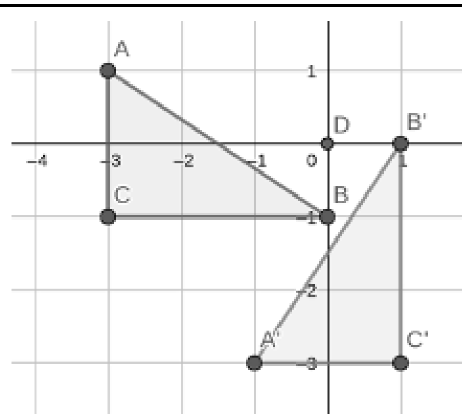


4. Sarah says it is a translation. Do you agree? Explain and provide the units of translation if you agree. I disagree because the orientations are not the same.

5. Diego says it is a rotation. Do you agree? Explain and provide the angle and direction of rotation.
I disagree because there was no spin.

6. Jayden says it is a reflection. Do you agree? Explain and provide the line of reflection if you agree.
I agree that it is a reflection because they are mirror images. The line of reflection is $y = -1$.

Describe each transformation. Include the units of translation, the line of reflection, or angle and direction of rotation as needed.

	Transformation		Description
7.	Quadrilateral $ABCD$ is the preimage and Quadrilateral $A'B'C'D'$ is the image.		Quadrilateral $ABCD$ was translated 2 units left and 1 unit down.
8.	$\triangle PQR$ is the preimage and $\triangle DEF$ is the image.		$\triangle PQR$ was rotated about the origin 90° counterclockwise or 270° clockwise.
9.	Quadrilateral $ABCD$ is the preimage and Quadrilateral $A'B'C'D'$ is the image.		Quadrilateral $ABCD$ was translated 2 units left and 3 units down.
10.	$\triangle ABC$is the preimage and $\triangle A'B'C'$ is the image.		$\triangle ABC$ was rotated about the origin 90° counterclockwise or 270° clockwise.